No of Pages : 3 Course Code : 23MX11

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, JUNE 2024

MCA Semester: 1

23MX11 MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks: 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Graph sheet is to be provided
3. Chi-Square table/t-table/Z-table are permitted to use
4. Subdivision (a) carries 3 marks each, subdivision (b) carries 7 marks each and subdivision (c) carries 15 marks each.

1. a) In a survey it was found that 21 people liked product A, 26 liked product B and 29 liked product C. If 14 people liked products A and B, 12 people liked products C and A, 14 people liked products B and C and 8 liked all the three products. Find how many liked product C only.

   b) i) Let P be the set of all people. Define a relation R on P by x R y iff x has the same eye color as y. Prove that R is an equivalence relation on P. [3]

   ii) If f is an invertible function from B to C, g is an invertible function from A to B. Prove that the inverse of the composition (f•g) is given by $(f \bullet g)^{-1} = g^{-1} \bullet f^{-1}$. [4]

   c) i) Determine the validity of the following argument. My father praises me only if I can be proud of myself. Either I do well in sports or I cannot be proud of myself. If I study hard, then I cannot do well in sports. Therefore, if father praises me, then I do not study well. [10]

   ii) Is $3^n - 1$ is multiple of 2? Validate the answer using Mathematical Induction. [5]

2. a) A company makes two kinds of leather belts, A and B. Belt A is high quality belt, and B is of lower quality. The respective profits are Rs 4 and Rs 3 per belt. Each belt of type A requires twice as much time as a belt of type B, and if all belts were of type B, the company could make 1000 belts per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle, and only 400 buckles per day are available. There are only 700 buckles available for belt B. Formulate the problem as a LPP

   b) Use graphical to solve the following LP problem.

   Maximize $z = 6x1 + x2$,
   Subject to: $2x1 + x2 \geq 3$,
   $x2 - x1 \geq 0$; $x1, x2 \geq 0$

   c) i) 1) Solve the following linear programming problem using **Simplex method** [10]

   Maximize $Z = 30x + 40y$
   Subject to: $2x + y \leq 10$;
   $x + y \leq 7$;
   $x + 2y \leq 12$; $x, y \geq 0$

2) Consider the problem of assigning five jobs to five persons. The assignment costs are given as follows. Determine the optimum assignment schedule.[5]

		Job				
		1	2	3	4	5
Person	A	8	4	2	6	1
	B	0	9	5	5	4
	C	3	8	9	2	6
	D	4	3	1	0	3
	E	9	5	8	9	5

(OR)

ii) Find the optimal solution of the following linear programming problem using **Big - M method**.

Maximize $Z = 5x1+12x2+4x3$

Subject to: $x1+2x2+x3 \leq 5$

$2x1-x2+3x3 = 2;\ x1, x2, x3 \geq 0$

3. a) A box contains cards numbered 1 to 30. A card is drawn from the box at random. What is the probability of getting a number on that card is an even number?

b) i) The chances of X, Y, Z becoming Managers are 4:2:3. The probability that Bonus scheme will be introduced in the company if X, Y, Z become Managers is 0.3, 0.7 and 0.8 respectively. If the Bonus scheme has been introduced, what is the probability that the A has been appointed as Manager? [4]

ii) Let X and Y be two random variables. The joint density function is given as f(X,Y) =kxy for 0<x<4 and 1<y<5; otherwise 0. Find k. [3]

c) i) 1) In Tim's office, there are 25 employees. Each employee travels to work every morning in his or her own car. The distribution of the driving times (in minutes) from home to work for the employees is shown in the table below. Calculate the mean of the driving times. [5]

Driving times	0-10	10-20	20-30	30-40	40-50
Number of Employees	3	10	6	4	2

2) The following frequency distribution table shows the monthly consumption of electricity of 68 consumers of a locality. Find the median of the given data. [10]

65-85	85-105	105-125	125-145	145-165	165-185	185-205
4	5	13	20	14	8	4

(OR)

ii) 1) Calculate the mean, variance and standard deviation for a random variable X defined as the number of tails in four tosses of a coin. Also, show the probability distribution. [5]

2) If the joint probability density function of a two-dimensional random variable (X, Y) is given by f(X,Y) = 8xy for 0<x<1 and 0<y<x; otherwise 0. Find: a) the marginal density functions of X and Y b) the conditional density functions and c) Check for independence of X and Y and P [Y < $\frac{1}{8}$ | X < $\frac{1}{2}$] [10]

4. a) What is the meaning of exponential smoothing in time series analysis?

b) A government agency reported that the population mean score on the quantitative portion of the Graduate Record Examination (GRE) General Test for students taking the exam between 2014 and 2017 was 558 ± 139 (μ ± σ). Suppose we select a sample of 100 students enrolled in a school with low funding and resources (n = 100). We hypothesize that students at this school will score lower than the general population. We record a sample mean equal to 585 (M = 585), same as measured in the above problem. Compute the one–independent sample z test [single tailed] at a 0.05 level of significance.

c) The following data lists the age and blood pressure [BP] of 10 sports persons.

Age[X]	42	36	55	58	35	65	60	50	48	51
BP [Y]	98	93	110	85	105	108	82	102	118	99

i) Find the regression equation Y on X and X on Y
ii) Find the correlation Coefficients [r] using the regression coefficients.
iii) Estimate the blood pressure of a sports person whose age is 45.

/END/

FD/RL